Less Is More: A Lightweight Deep Learning Network for Remote Sensing Imagery Segmentation

Jianwei Yue, Yan Wang, Jie Pan, Haojian Liang, Shaohua Wang, and Quanyi Liu

***Abstract*— High-resolution (HR) remote sensing (RS) imageries contain a wealth of information about ground object features. Semantic segmentation of HR RS imageries has emerged as a prominent area of research. However, many existing models have become complex in pursuit of accuracy, making it challenging to achieve a balance between efficiency and precision. To address this issue, we propose LiteSeger, a lightweight semantic segmentation model that combines the strengths of convolutional neural network (CNN) and Transformer architectures. Initially, the MobileViT backbone network, which integrates CNN and Transformer, is employed to rapidly extract multilevel features. Subsequently, we devised a global attention mechanism and a local attention mechanism, respectively, for low- and high-level features, which we incorporated into the decoder as the main feature fusion (MFF) module. Local attention enhances the representation of detailed features in both spatial and channel dimensions. Global attention strengthens global interconnectivity through long-range semantic information. In particular, our proposed distance-aware grouped attention demonstrates a significant reduction in the computational complexity of attention computation, without compromising segmentation accuracy. In comparison to other lightweight models, our proposed LiteSeger achieves the best mean intersection over union (MIoU) and average $F1$-score (Ave.F1) on both the Vaihingen and Potsdam datasets. On the Vaihingen dataset, it achieves an MIoU of 71.22 and an Ave.F1 of 82.97. On the Potsdam dataset, the MIoU and Ave.F1 reach 72.00 and 83.34, respectively. At the same time, it has significant advantages in inference efficiency. LiteSeger contains only 7.0 M parameters, and under the experimental settings of this study, it achieves an inference speed of 50 FPS. The source codes are available at https://github.com/HIGISX/LiteSeger**

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I. Introduction

THE rapid advancement of remote sensing (RS) technology has resulted in the acquisition of a considerable number of RS data, e.g., high resolution (HR) [1], hyperspectral (HS) [2], and light detection and ranging (LiDAR) [3], which have become a valuable resource for researchers in various fields [4]. These RS imageries are of significant value for the study of Earth sciences, which has become a primary focus of research aimed at understanding the knowledge contained in these imageries and efficiently filtering the information of interest [5], [6], [7]. The high-resolution RS imageries are replete with detailed spatial information, which can be utilized for the classification of land cover and land use [8]. Conventional approaches to land cover and land use classification, such as visual interpretation [9] and certain machine learning techniques (e.g., SVM [10], [11], random forest [12], and decision tree [13]), typically require a significant investment of human resources and are not particularly efficient. This hinders the potential for utilizing high-resolution RS imageries.

In recent years, deep learning methods have provided methodological support for land cover and land use classification tasks of high-resolution RS imageries, particularly in the context of semantic segmentation tasks in computer vision. The convolutional neural network (CNN) is a widely studied and applied as a classical model for deep learning methods, with the fully convolutional network (FCN) [14] playing a crucial role. The encoder–decoder structure has emerged as the dominant model for segmentation tasks, as evidenced by the extensive research conducted by numerous scholars, due to its superior performance in segmentation. A notable example is the U-Net model [15], which learns both the low-level spatial features and the high-level semantic features of the input images layer-by-layer through the encoder. Subsequently, it integrates multiscale features using skip connections and performs fine upsampling in the decoder, thereby restoring the original image size. A number of CNN-based semantic segmentation methods have emerged in the field of RS [16], [17], [18], [19], [20]. However, CNN networks have been observed to lose small-scale features during downsampling [21], which has been demonstrated to degrade the segmentation performance. Standard convolution operations are constrained in their capacity to interactively learn global contextual information due to their focus on local information proximate to the receptive field [22]. Furthermore, there are

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considerable challenges posed by the substantial variance in object scale, high interclass similarity, and mutual occlusion in RS imageries [23] (Fig. 1).

The advent of the Vision Transformer (ViT) [24] has ushered in a novel approach to modeling global relationships in imageries. The Transformer structure divides the original image into small size patches (e.g., 8×8 and 16×16) and models the correlation between each patch using a multihead self-attention mechanism to achieve global interoperability of information. Carion et al. [25] achieved efficient completion of relevant tasks in an encoder–decoder structure by combining CNN feature extraction with Transformer global information modeling. T2T-ViT [26] reduces the training difficulty of ViT by designing a "Token-to-Token" approach. In the field of RS imagery segmentation, a number of approaches have been developed that combine CNN and Transformer, with the objective of further enhancing the segmentation efficacy. Examples of such approaches include ST-Unet [27], UNetFormer [28], and CTFuse [29].

Although progress has been made in the field of segmentation results, the number of network parameters and the associated computational complexity have undergone a notable surge. This has led to an increased demand for substantial computational resources and prolonged training times. Consequently, a number of lightweight network architectures have been developed. BiseNet [30] achieves a lightweight model architecture by designing a spatial path and a context path, which combine spatial correlation and channel attention. By utilizing linear attention mechanisms (LAMs), MAResU-Net [31] reconstructs the skip-connection layers of U-Net, balancing the acquisition of global information with efficiency. However, lightweight networks typically have shallow layers, which limits the features that can be learned, making it challenging to strike a balance between efficiency and accuracy.

In this study, we present novel lightweight deep learning network structures for the field of RS imagery segmentation. The objective is to enhance the accuracy of the segmentation while maintaining efficiency. To this end, an encoder–decoder structure is employed, whereby MobileViT [32] is used as the backbone. This enables the rapid and effective capture of both local features and global information. Furthermore, the features generated during the encoding stage are integrated into the decoding stage, thereby enhancing the segmentation performance of the network model while only minimally increasing the number of parameters. The main contributions of this study are as follows.

1) We design a lightweight deep learning framework for high-precision RS imagery segmentation. By employing a lightweight backbone network, the primary features of the original image are rapidly discerned. By integrating the Transformer and CNN structures into a designed decoder, the loss of main object information and edge details during the decoding stage is reduced, thereby further improving the segmentation accuracy of the proposed network.
2) A global–local decoder structure, main feature fusion (MFF), is proposed. By designing the local attention and global attention structures, fine-grained global features and enhanced local features are fused to efficiently model the global contextual features of the image and enhance the overall segmentation effect.
3) We propose distance-aware grouped attention, an efficient attention mechanism designed to reduce computational complexity while preserving fine-grained spatial details. It achieves efficient long-range similarity computation while preserving fine-grained details through its innovative group-based patch partitioning strategy.



Fig. 1. Samples of RS imageries from the Vaihingen dataset are shown. (a) Shadow interference is demonstrated. (b) Interference of occlusion on feature extraction. (c) High similarity between different categories.

II. Related Work

RS imagery segmentation methods play a crucial role in land cover and land use tasks, which are widely applied in urban planning, agricultural land change analysis, and other domains. In recent years, with the rapid development of deep learning methods, the accuracy of multiclass segmentation in RS imageries has reached a new level. This section will introduce a few pivotal deep learning methods related to RS segmentation and discuss their impact and contributions to the field.

A. CNN- and Transformer-Based RS Image Segmentation

Since the application of deep learning methods in the field of RS imagery segmentation, deep learning networks have been able to learn richer image features due to their multilevel hierarchical structure. By nonlinear transformations between different layers, the extraction accuracy of the model is substantially improved. Currently, RS imagery segmentation methods are usually CNN-based and Transformer-based.

Among the methods for CNN-based RS imagery segmentation, U-Net [15] has gradually become a standard model due to its high accuracy performance. Dong et al. [33] achieved the aggregation of small-scale features through a dense fusion strategy and designed DenseU-Net to effectively improve the accuracy and overall accuracy of small object classes. RefineUNet proposed by Qiu et al. [34] improves the skip-connection scheme in network architecture. Due to the weak ability of

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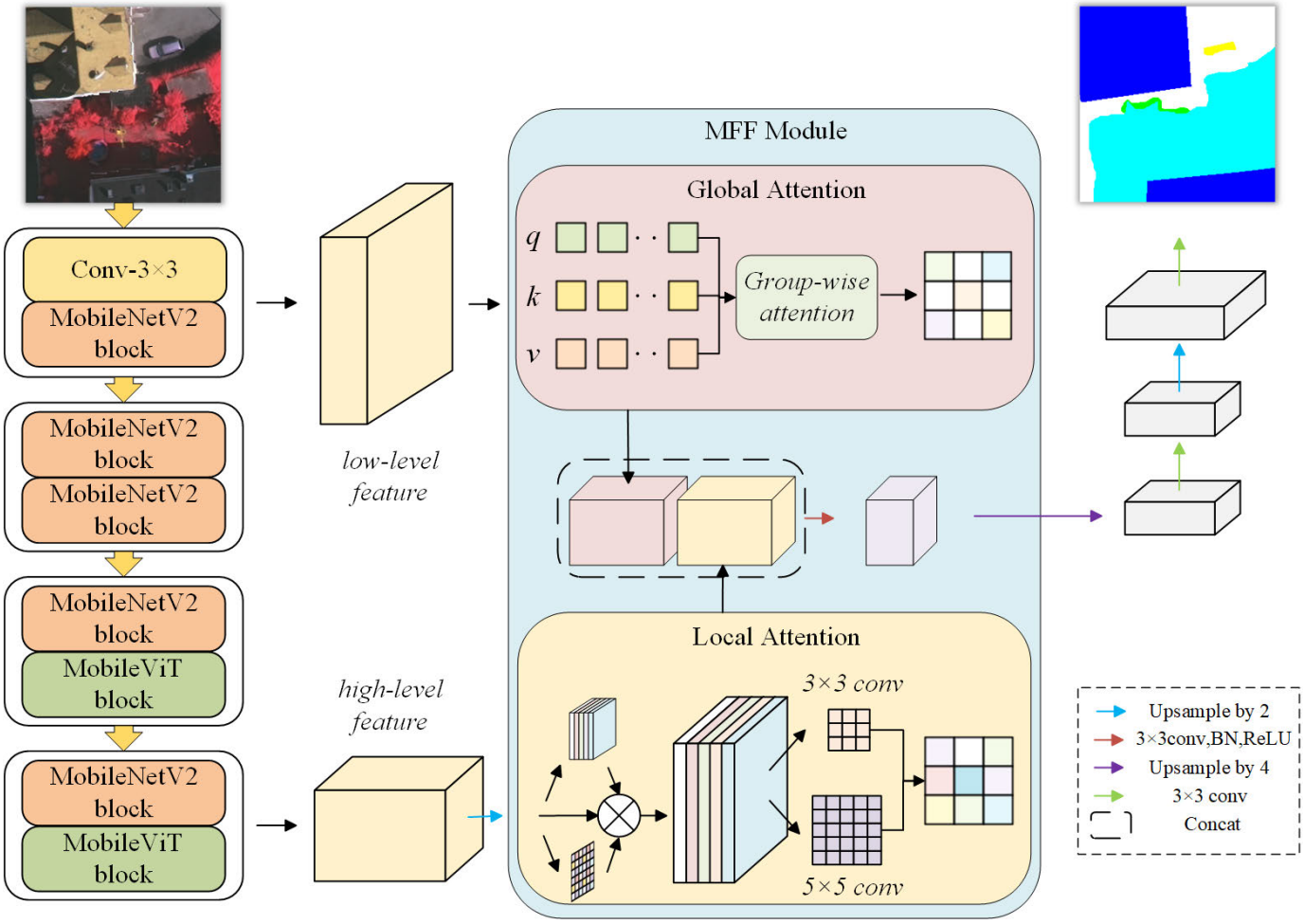


Fig. 2. Architecture of our proposed LiteSeger.

convolutional operations to capture long-distance dependencies, some researchers have focused on the global relationships between objects in images and introduced attention mechanisms into their models [5], [35], [36]. Hong et al. [37] recognized the advantage of graph convolutional networks (GCNs) in modeling long-range relationships and explored strategies for integrating GCNs and CNNs to extract imagery features. There are also some approaches based on other frameworks, e.g., DeeplabV3+ [38] and FCN [14].

The ViT [24] opens up the application of Transformer [39] structures in the image domain. SETR [40] and Segmenter [41] directly use the traditional Transformer sequence-to-sequence pattern for image segmentation. Wang et al. [42] used Swin Transformer [43] as the backbone to leverage contextual information. In contrast to CNN-based methods, these Transformer-based methods adequately capture the global relationship of image patches. However, the global modeling characteristics of Transformer significantly increase the complexity of the algorithms, which makes it more difficult to train.

There are also studies that combine CNNs and Transformers. ST-UNet [27] used Swin Transformer as an auxiliary encoder for feature extraction, while CTFuse [29] used convolution in the first two stages of encoding, and stages 3 and 4 used the Transformer structure. UNetFormer [28], on the other hand, fused CNN and Transformer in the decoding stage. While the models above have achieved significant improvements in accuracy, the increased model parameters and model complexity make model training require more computational resources, and predictions consume more inference time and are more difficult to deploy for real-time feedback applications.

In this study, we combine the structural characteristics of CNN and Transformer to design a new module to improve the efficiency of the model and ensure the accuracy.

B. Lightweight Model for RS Image Segmentation

With the development of the Internet and mobile communications, the demand for mobile-based devices is surging, and a series of lightweight deep learning network architectures have emerged, such as ResNet-18 [44], the MobileNet series [45], [46], and ShuffleNet [47]. These networks are designed to reduce the number of parameters and computational complexity of the models, allowing them to operate efficiently even in resource-constrained environments.

For very-high-resolution (VHR) RS imagery, which contains rich spatial and spectral details, lightweight models are particularly advantageous in tasks such as land cover

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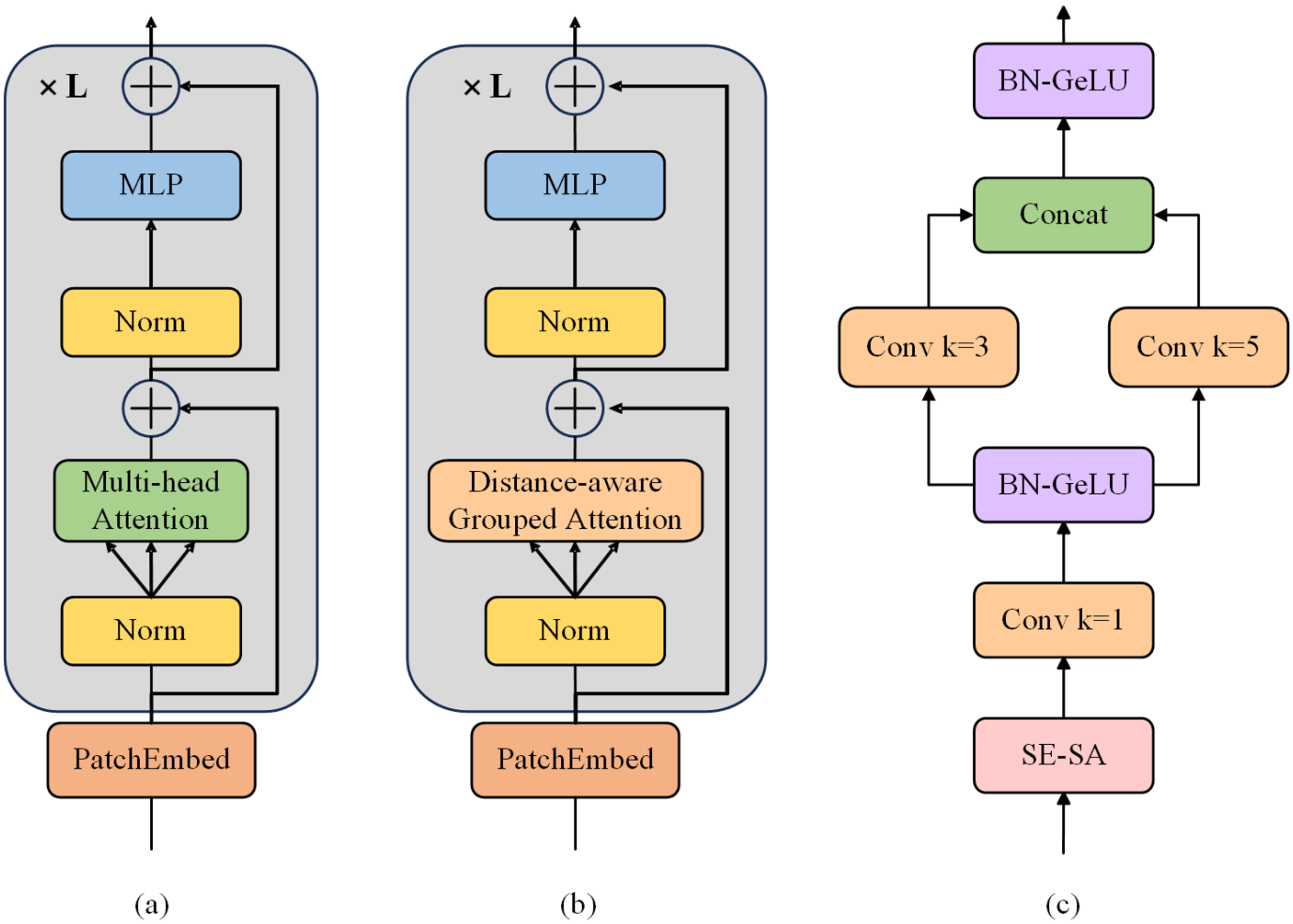


Fig. 3. Structure of global attention and local attention in the module MFF of LiteSeger. (a) Structure of standard Transformer block, which uses multihead attention to learn global features. However, multihead attention takes all sequence length of input patches N, which makes its computational complexity $O(N^2)$. (b) Structure of global attention, which aims to enhance the representation of long-distance information correlations by utilizing the fine-grained features of the low-level features. Compared to the standard Transformer block, we replace the multihead attention with our proposed distance-aware grouped attention. Distance-aware grouped attention groups patches by a size of g, which effectively reduces the complexity to $O(N^2/g^2)$. (c) Structure of LOCAL ATTENTION, which is a module that first enhances global relevance through SE-SA and then fuses local information through small-kernel convolution to improve the extraction quality of main features.

classification and urban monitoring. These models mitigate the computational burden associated with processing high-resolution images, enabling real-time applications such as dynamic land use surveys and environmental monitoring [48]. For instance, Li et al. [8] proposed an LAM that reduces the memory and computational cost of the traditional dot-product attention mechanism. Similarly, Wang et al. [49] designed the bilateral awareness structure to ensure the efficiency while taking into account the long-distance relationships and fine-grained details in the imagery. With large kernel convolution and a spatial kernel selection mechanism, LSKNet [50] can efficiently learn the multiscale features in RS imageries.

Despite these advancements, lightweight models often face a tradeoff between efficiency and accuracy. To address this, recent studies have explored hybrid architectures that combine the strengths of CNNs and Transformers. Ji et al. [51] developed E-Net, an efficient and lightweight CNN with a novel E-shaped architecture and a contextual aggregation attention mechanism (CA2M), achieving state-of-the-art performance in sea–land segmentation with minimal computational cost. Qiu et al. [52] proposed GATrans, a generative adversarial Transformer that leverages a global Transformer network to efficiently extract multilevel features while maintaining a compact model size. Chen et al. [53] presented LEFormer for lake boundary extraction, which integrates CNN encoders, Transformer encoders, and a cross-attention fusion module to enhance segmentation performance. Ge et al. [54] proposed CD-CTFM, a lightweight CNN-Transformer fusion model that enhances multiscale feature representation with a pyramidal architecture. These approaches demonstrate that carefully designed lightweight models can achieve high accuracy without sacrificing efficiency.

In this study, based on the lightweight backbone, a series of feature fusion modules and a new attention mechanism are designed to combine the advantages of convolution and Transformer to improve the imagery segmentation accuracy while ensuring the efficiency.

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III. Method

This section will first present the general structure of the proposed LiteSeger. It will then analyze the principles and roles of the global attention and local attention branches within it, specifically with regard to the designed MFF module.

A. Network Structure

The overall architecture of our LiteSeger is shown in Fig. 2. The network structure can be divided into two parts: backbone network encoder and MFF module, SegHead decoder.

For the backbone network, we choose MobileViT [32] as the encoder. On the one hand, MobileViT is a lightweight network structure that can quickly acquire features at different levels of the imagery and maintain a high speed in the inference phase. On the other hand, it effectively combines CNNs and Transformers. Convolution is performed in the first two stages to obtain the main features in the imagery to aggregate effective information from the local receptive field. The last two stages use Transformer to model global information and enhance the correlation between the same category of features over long distances.

For a given RS imagery $X \in \mathbb{R}^{H\times W\times 3}$, after encoding by the encoder, a high-level feature $X_h \in \mathbb{R}^{(H/16)\times(W/16)\times 4C}$ and a low-level feature $X_l \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ are obtained (the value of C depends on the specific channel number setting of the backbone network).

After that, X_l and X_h are fed into the MFF module for global attention and local attention processing, respectively. The MFF module is used to augment the high-level feature and low-level feature, fusing the features to enhance the overall effect of segmentation. In order to model the global features on a larger scale, the low-level feature is processed by global attention to obtain $X_l' \in \mathbb{R}^{(H/8)\times(W/8)\times 8C}$. For high-level features, global relevance is enhanced by local attention and main features are aggregated in small receptive fields to reduce the omission of small-scale target features. The obtained $X_h' \in \mathbb{R}^{(H/8)\times(W/8)\times 8C}$. Concatenate X_l' and X_h'. Then, a dimensionality reduction operation is performed through a 3×3 convolutional layer to reduce the size to keep it lightweight and upsample to $(H/2) \times (W/2)$ size output. Finally, the features are upsampled to $H \times W$ size by SegHead to get the prediction mask.

B. MFF Module

As illustrated in Figs. 2 and 3, the MFF module comprises three principal components: global attention, local attention, and fusion part. The following subsections provide a detailed description of these components.

1) Local Attention: The output features from the encoder have exhibited robust global correlations, yet concurrently, there has been a discernible diminution in local information. Local attention addresses the aforementioned issue by initially reinforcing the global correlation and subsequently enhancing the aggregation of local information. Inspired by [55], SE-SA reinforces global relevance in terms of channel and space (Fig. 4). Squeeze excitation (SE) first spatially compresses the features and computes weights for each channel through the inverted bottleneck structure. Compared to Transformer, spatial attention (SA) interacts spatially at the pixel scale and refines the connection between long-distance features more. Afterward, the inverted bottleneck is formed using convolution kernels of sizes 1×1, 3×3, and 5×5 to enhance the expression of local information and reduce the risk of small-scale feature loss.

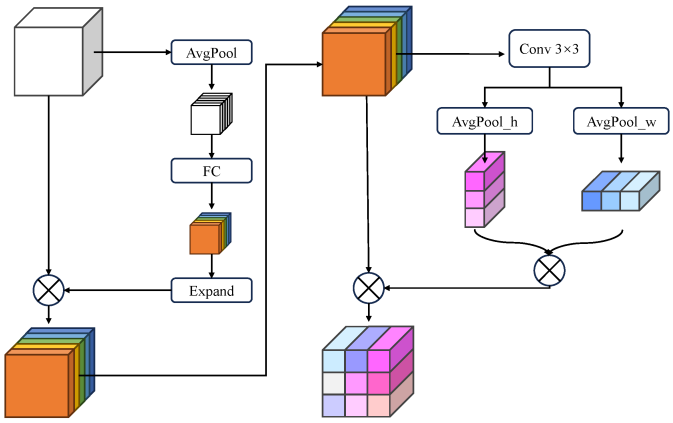


Fig. 4. Overall structure of SE-SA. The SE module in the left half aims to suppress useless information expression from the channel, and the SA module in the right half enhances the spatial correlation at the pixel level.

The equation for SE is expressed as follows:

$$z_c = \frac{1}{h \times w}\sum_{i=1}^{h}\sum_{j=1}^{w} x_c(i, j) \tag{1}$$

$$\tilde{x} = x \otimes \sigma(W_2\delta(W_1 z)) \tag{2}$$

where h and w represent the height and width of the input feature tensor, respectively, and $x_c(i, j)$ represents the feature pixel at the (i, j) position. Through 1, we get the pooling value of channel c. W_1 and W_2 represent linear projection for inverted bottleneck structure. $\delta(\cdot)$ stands for ReLU activation function and $\sigma(\cdot)$ stands for sigmoid activation function. $\otimes$ means the matrix multiplication.

The equation for SA is expressed as follows:

$$z_{w_j} = \frac{1}{h}\sum_{i=1}^{h}\tilde{x}_c(i, j) \tag{3}$$

$$z_{h_i} = \frac{1}{w}\sum_{j=1}^{w}\tilde{x}_c(i, j) \tag{4}$$

$$x_{\text{out}} = \tilde{x} \otimes (z_h \otimes z_w). \tag{5}$$

Using (3) and (4), we obtain the aggregated values of the input tensor in the horizontal and vertical directions, denoted as z_{h_i} and z_{w_j}, respectively. Then, z_h and z_w are multiplied to obtain the spatial weights.

2) Global Attention: The low-level feature still retains rich detail features that would be destroyed if a convolutional operation were used (e.g., house edges and cars) The global features constructed by the MobileViT block in the encoder are lacking sufficient location information; therefore, the global attention block aims to establish the connection between long-distance semantic information from a more detailed point

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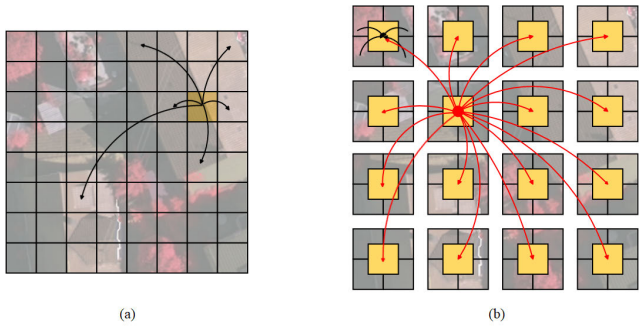


Fig. 5. Comparison of the conventional attention and distance-aware grouped attention. (a) Conventional attention mechanism computes self-attention between every pair of patches. However, in RS images, adjacent patches often belong to the same category. In practice, we want the attention mechanism to focus more on long-range dependencies between distant patches. Performing global self-attention for all patches would introduce significant computational redundancy. (b) Distance-aware grouped attention employs learnable weights to group adjacent patches, which not only preserves the positional information of each patch but also effectively reduces computational complexity when performing self-attention.

of view and improve the ability to recover location information. As shown in Fig. 3(b), the global attention block uses our proposed improved ViT block. Conventional ViT blocks typically perform patch embedding with relatively large sizes (e.g., 7×7 and 8×8). However, in RS imageries, the size of objects varies greatly. Large-sized patches can lead to the loss of a significant number of fine details and positional information, resulting in blurred edges in the segmented objects. While smaller patches preserve fine-grained spatial details, they inevitably increase the sequence length, thereby inducing higher computational complexity. To preserve detailed information while maintaining computational efficiency, we replace the standard multihead attention with our proposed distance-aware grouped attention. As illustrated in Fig. 5(b), the proposed method first groups adjacent patches into local clusters for feature aggregation. The similarity computation is then performed on these aggregated group features, which simultaneously captures long-range dependencies while significantly reducing computational complexity.

C. Distance-Aware Grouped Attention

The conventional self-attention performs pairwise similarity computation among all patches. Given an input tensor $X \in \mathbb{R}^{N \times d}$, where N denotes the sequence length and d represents the feature dimension, the computational complexity of standard self-attention scales quadratically as $O(N^2)$, indicating that the computational complexity is determined by the squared length of the patch sequence. As discussed previously, preserving finer spatial details will substantially increase the number of patches (N), consequently leading to an exponential growth in computational complexity.

Inspired by the window-based design of Swin Transformer [43], we propose an innovative grouping mechanism to balance computational efficiency and feature representation. As illustrated in Fig. 6, we partition the patches sequence into $G = N/g$ nonoverlapping groups, each containing g consecutive patches. For each group, we compute a shared

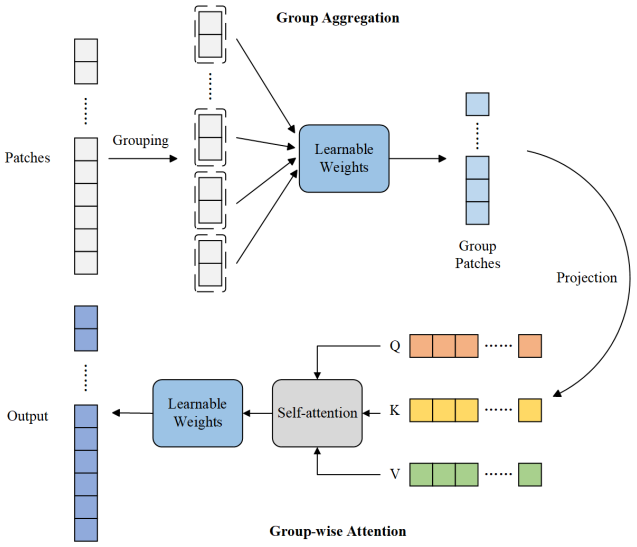


Fig. 6. Overall structure of distance-aware grouped attention. Patch tokens are grouped and aggregated with learnable weights, followed by group-level attention. The attended representations are then broadcast back to the original patch space.

learnable weighted sum to obtain a single group patch F_i

$$F_i = \sum_{j=1}^{g} w_j x(i, j) \tag{6}$$

where w_j is the learnable parameter shared across all groups and updated during training and $\sum_{j=1}^{g} w_j = 1$.

With the reduced patch sequence $F \in \mathbb{R}^{G \times d}$, we apply self-attention across group patches

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{7}$$

where $Q, K, V \in \mathbb{R}^{G \times d}$. This captures long-range dependencies between spatial regions while significantly reducing computation. The number of attention score computations drops from N^2 to G^2. This modification reduces the computational complexity to $(1/g^2)$ of the original implementation. Considering both detailed feature representation and computational complexity, we empirically set the group size g to 4.

To restore the group-level attention back to the token space, we use weighted reprojection. The learnable weights not only aggregate patch representations but also inherently preserve the positional information of original patches. This significantly facilitates the reprojection from the attention space back to the original patch space. The output X_i'

$$X_i' = F_i' w_i. \tag{8}$$

Our method retains the flat ViT structure and introduces learnable, data-driven aggregation that preserves patch details while improving efficiency.

IV. Experiments

A. Datasets

The Vaihingen and Potsdam datasets are widely utilized in RS semantic segmentation tasks and are now regarded as a standard data source in the field. Due to the extensive research

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and use of these datasets, it is possible to conduct a more comprehensive comparison of our work with previous studies. The following is a detailed description of the two datasets.

1) Vaihingen Dataset: The dataset contains 33 high-resolution TOP imageries, each with an average size of 2494 × 2064 pixels. The dataset covers 1.33 km^2 of the Vaihingen area. Each imagery consisted of three channels, infrared (IR), red (R), and green (G), and had a ground sampling distance (GSD) of 9 cm. The dataset contains a total of five foreground categories (impervious surfaces, buildings, low vegetation, trees, and cars) and one background category (clutter). In order to ensure the randomness of the dataset division to enhance the generalization ability of the model, we crop the imageries in the dataset to 256 × 256 pixels and randomly divide them into the training set and the test set in the ratio of 8:1.

2) Potsdam Dataset: The dataset contains 38 high-resolution TOP imageries, each of which has a size of 6000 × 6000 pixels. The dataset covers an area of 3.42 km^2 in the Potsdam region with a GSD of 5 cm. There are three channel combinations in this dataset, and in this article, the channel combinations of red (R), green (G), and blue (B) are chosen to carry out the experiments and accuracy evaluation. Considering the size of the dataset, we crop the imageries to 256 × 256 size and divide the training set and test set according to the ratio of 4:1. More test imageries can better verify the segmentation ability of the model.

B. Implementation Details

1) Training Settings: This study implemented the network modeling based on the Pytorch 1.9 framework. Experiments were conducted using NVIDIA RTX A3000 12-GB GPUs. The SGD algorithm with momentum has been proven to optimize models more effectively [56], [57]. Therefore, this study used an SGD optimizer with a momentum item of 0.9 and a weight decay of 1×10^{-4} to train the model. The initial learning rate was set to 0.01, the batch size was set to 8, and the maximum epoch was set to 100.

2) Loss Function: It has been explained in [27] and [58] that the Vaihingen and Potsdam datasets have a category imbalance problem that affects the effectiveness of model training for the categories in a small percentage. Therefore, the loss function used in this article is a combination of Dice loss and cross-entropy loss function, which can be expressed as

$$L_{\text{ce}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{9}$$

$$L_{\text{dice}} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)} y_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \tag{10}$$

$$L = L_{\text{ce}} + L_{\text{dice}} \tag{11}$$

where N represents the number of samples; K represents the number of classes; and $y^{(n)}$ and $\hat{y}^{(n)}$ represent the true label and its corresponding predicted output, respectively.

3) Evaluation Metrics: In this experiment, the metrics commonly used in RS segmentation tasks are employed: mean intersection over union (MIoU) and average $F1$-score (Ave.F1). These two evaluation metrics are based on the confusion matrix, which contains four items: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). The specific equation for IoU is as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{12}$$

The $F1$-score of each category is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{15}$$

MIoU denotes the average of IoU for all categories and Ave.F1 denotes the average of $F1$-score for all categories.

To evaluate the size and efficiency of the model, we have also introduced two additional metrics: parameters and speed.

TABLE I
ABLATION RESULT OF DIFFERENT LOSS FUNCTION

Loss	Vaihingen		Potsdam	
	MIoU(%)	Ave.F1(%)	MIoU(%)	Ave.F1(%)
CE	70.51	82.45	71.45	83.19
CE+Dice	71.22	82.97	72.00	83.34

C. Ablation Study

To evaluate the performance of the different components of LiteSeger, we conducted a series of ablation experiments on the Vaihingen dataset. In addition, we analyzed the impact of different architectural compositions on the overall performance of the network.

1) Effect of Loss Function: The experimental results with different loss functions are presented in Table I. After applying the composite loss function, the experimental results demonstrate significant accuracy improvements across both datasets. This shows that Dice loss effectively addresses class imbalance and enhances model training performance.

2) Effect of Local Attention: The results of the ablation experiments for different components are shown in Table II. When using the MobileViT as a backbone network, joining the LA components MIoU and Ave.F1 improves 0.84% and 0.61%, respectively. The "Car" category improved the most, by 2.86%, followed by "Imp.Sur," by 0.68%. In terms of the visual segmentation results, as shown in Fig. 7, the edges of the cars in the first row (a) are chaotically attached together and (b) the use of LA is able to segment the complete edges of the individuals from the tightly packed group of cars. In the second row, at shaded building edges, the use of LA components attenuates the negative impact of shadows on the segmentation results. Experiments show that LA is able to better aggregate similar features in a small-scale range, which improves the segmentation accuracy of feature edges.

3) Effect of SE Block and SA Block: The experimental results of removing the SE block and SA block in the LA component are shown in Table II. Compared to using the full LA component, removing the SE block results in a loss of 0.16% for

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TABLE II
ABLATION RESULT OF THE DIFFERENT COMPONENTS ON THE VAIHINGEN DATASET

Method	IoU (%)					Evaluation Index	
	Imp.Sur.	Building	Low Veg.	Tree	Car	MIoU (%)	Ave.F1 (%)
MobileViT	71.69	82.78	61.23	70.79	63.22	69.94	82.08
MobileViT+LA	72.37	82.52	61.48	71.47	**66.08**	70.78	82.69
MobileViT+LA w/o SE	72.36	82.44	61.72	71.37	65.20	70.62	82.58
MobileViT+LA w/o SA	72.61	83.29	**62.59**	70.33	64.03	70.57	82.53
MobileViT+GA	**73.41**	82.54	61.86	**72.73**	64.53	71.01	82.95
LiteSeger	72.64	**83.62**	62.33	72.31	65.20	**71.22**	**82.97**

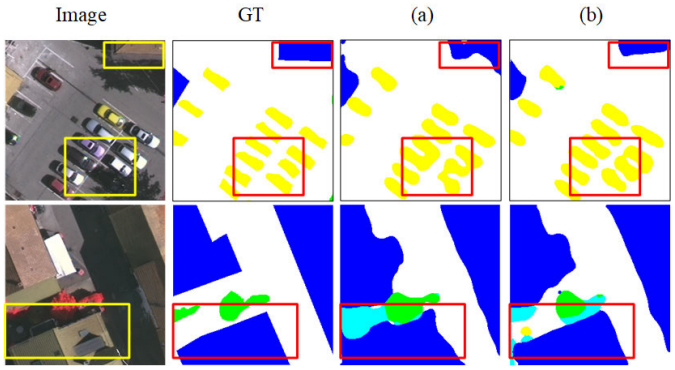


Fig. 7. Comparison of segmentation results before and after using LA in our framework. (a) MobileViT. (b) MobileViT + LA.

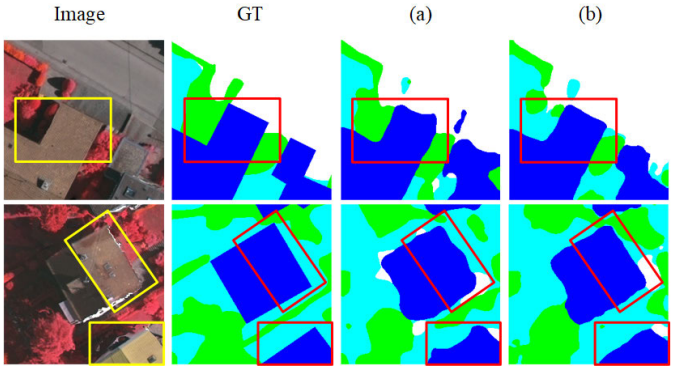


Fig. 8. Comparison of segmentation results before and after using GA in our framework. (a) MobileViT. (b) MobileViT + GA.

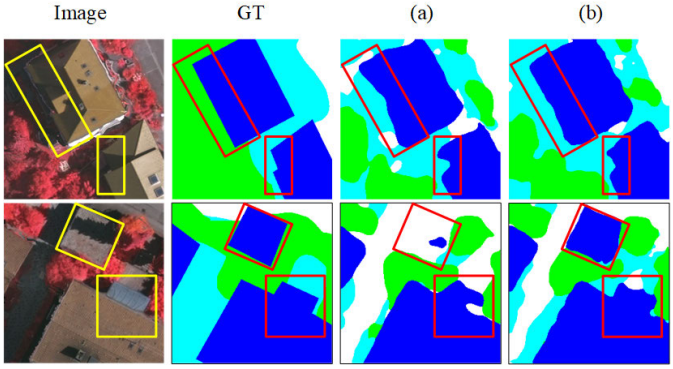


Fig. 9. Comparison of segmentation results before and after using MFF in our framework. (a) MobileViT. (b) MobileViT + MFF (LiteSeger).

MIoU and 0.11% for Ave.F1, with the largest loss in the "Car" category, which is reduced by 0.88%. After removing the SA block, MIoU and Ave.F1 lose 0.21% and 0.16%, respectively, with the "Car" and "Tree" categories losing 2.05% and 1.14%, respectively. The experimental results show that the addition of both SE block and SA block improves the overall segmentation quality of LA and makes the segmentation results of each category more balanced.

4) Effect of Gocal Attention: Table II shows that after using the GA component, the segmentation results on MIoU and Ave.F1 increased by 1.07% and 0.87%, respectively. The IoU of the "Imp.Sur." and "Tree" categories increased by 1.72% and 1.94%, respectively. The edges of "Building" in RS imageries are usually mixed with "Low Vegetation," and the blurred demarcation line and shadows make the segmentation difficult. As can be seen in Fig. 8, after the introduction of GA, the edge segmentation of building corners is significantly improved, effectively restoring the detailed features of the objects. Simultaneously, the method enhances the separability between low vegetation and tree, yielding significantly improved segmentation accuracy. The experimental results show that GA has an enhanced effect on accurately identifying the position information of the object and recovering the main features.

5) Effect of MFF: When the MFF module is added (i.e., the LiteSeger structure), the results in Table II show an improvement of 1.28% and 0.89% on MIoU and Ave.F1, respectively. Specifically, the individual IoU increased in all five categories, with the "Low Vegetation" category improving by 1.10%, the "Tree" category improving by 1.52%, and the "Car" category by 1.98%. As shown in Fig. 9, the first row demonstrates the ability of LiteSeger for exact recognition of building edges. The second row shows its strong resistance to shadow interference. Based on the comprehensive experimental results, the MFF module can significantly improve the ability of the model to resist shadow interference and significantly improve the segmentation accuracy of various types of features.

6) Effect of Architecture: To explore the effect of the backbone on model performance, we conducted experiments on the Vaihingen dataset by replacing the backbone with different lightweight models. These backbones include ResNet18, MobileNetV2, MobileNetV4-s, and MobileViT-s. The experimental results are shown in Table III. The results indicate that when MobileViT-s is used as the backbone, both MIoU and Ave.F1 achieve the highest values, resulting in the best experimental accuracy. Therefore, we prefer MobileViT-s as the backbone.

To explore the optimal configuration of the model, we conducted combination experiments on the number of

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TABLE III
ABLATION RESULT OF DIFFERENT BACKBONES ON THE VAIHINGEN DATASET

Method	Backbone	MIoU (%)	Ave.F1 (%)
LiteSeger	ResNet18	70.54	81.67
	MobileNetV2	71.05	82.85
	MobileNetV4-s	70.21	81.54
	MobileViT-s	**71.22**	**82.97**

TABLE IV
ABLATION RESULT OF DIFFERENT CONFIGURATIONS ON THE VAIHINGEN DATASET

Method	$\{n, k1, k2\}^*$	MIoU (%)	Ave.F1 (%)
LiteSeger	{4, 1, 3}	70.71	82.63
	{4, 3, 5}	71.04	82.83
	{8, 1, 3}	70.75	82.63
	{8, 3, 5}	**71.22**	**82.97**

* n represents the number of transformer blocks in the GA module, $k1, k2$ represents the kernel size of the parallel convolution kernel in the LA module.

Transformer blocks in the GA module and the convolution kernel size in the LA module. The experimental results are shown in Table IV. The experimental results show that when the combination is {8, 3, 5}, the model achieves the best results in both MIoU and Ave.F1. Therefore, we prefer this design combination as the network's structural configuration of the network.

D. Comparison With Other Methods

In this section, we evaluate the proposed LiteSeger in comparison with a number of existing approaches, including the classical DeeplabV3+ [38], ResUnet, as well as a selection of lightweight models, namely, MAResUNet [31], ABCNet [8], BANet [49], UNetformer [28], LSKNet [50], CREVNet [59], and LiViTNet [60]. Additionally, we include a number of state-of-the-art heavyweight models, such as SETR [40], TransUnet [61], ST-Unet [27], and DC-Swin [42], in order to provide a comprehensive comparison. To ensure the comparability of the results, all of the aforementioned networks utilize the same training strategy and are not pretrained.

1) Result on Vaihingen Dataset: Table V lists the experimental results of each method on the Vaihingen dataset. The data in the table show that the proposed LiteSeger outperforms other methods in both MIoU and Ave.F1, with scores of 71.22% and 82.97%, respectively. Compared to models based on CNN methods, LiteSeger not only achieves higher accuracy than the purely convolutional ABCNet and the attention-enhanced MAResUNet but also outperforms the latest LSKNet, which utilizes large kernels. LiteSeger improves upon LSKNet by 3.19% in MIoU and 2.33% in Ave.F1. BANet introduces a bilateral awareness path, encoding with both convolution blocks and efficient Transformer blocks, while UNetFormer combines convolution and Transformer designs during the decoding stage. Compared to these hybrid models, LiteSeger outperforms BANet by 4.34% and UNetFormer by 1.18% in MIoU, demonstrating that LiteSeger's structural design offers a significant advantage in segmentation results. CREVNet and LiViTNet use the same backbone network as our method. In terms of MIoU and Ave.F1 scores, LiteSeger outperforms CREVNet by 12.54% and 9.24% and surpasses LiViTNet by 0.85% and 0.58%, respectively. This demonstrates that our proposed MFF architecture achieves superior segmentation accuracy and exhibits enhanced adaptability to VHR RS imagery. LiteSeger even surpasses some state-of-the-art heavyweight models, achieving 1.75% higher MIoU and 1.39% higher Ave.F1 compared to DC-Swin.

Fig. 10 illustrates the segmentation prediction outcomes of a number of methodologies presented in Table V. In the initial two rows, alternative methodologies fail to effectively integrate global and local information, resulting in the generation of numerous gaps in the "Building" segmentation outcomes. CREVNet, LiViTNet, and UnetFormer lose the detailed boundary of "Car." Our LiteSeger demonstrates robust capability in segmenting building contours and car edges within shadowed regions. In the subsequent two rows, the influence of shadow interference led to the misclassification of certain "Tree" segmentation regions as "Low Vegetation," whereas our LiteSeger exhibited comparatively precise predictions. In the outcomes of the fifth row, several methods (e.g., CREVNet and MAResUNet) suffer from loss of fine details in shadowed areas, particularly for buildings and impervious surfaces. In this comparative group, the hybrid architecture demonstrates superior segmentation performance. LiteSeger demonstrated superior performance in both object boundary segmentation and object classification.

2) Result on Potsdam Dataset: The experimental results of LiteSeger and comparison methods on the Potsdam dataset are shown in Table VI. In terms of MIoU, LiteSeger outperforms other lightweight methods and achieves the best segmentation IoU for the "Low Vegetation" class and second best for the "Impervious Surface," "Tree," and "Car" classes. Comprehensively, CNN-based methods exhibit superior performance in single-class segmentation tasks, whereas CNN-Transformer hybrid architectures achieve higher overall accuracy. Compared to the large-kernel convolution-based LSKNet and the CNN–Transformer hybrid UNetFormer, LiteSeger improves MIoU by 4.84% and 2.70%, respectively. Compared to CREVNet and LiViTNet with identical backbones, LiteSeger achieves superior performance with relative improvements of 16.40% and 0.90% in MIoU, and 12.05% and 0.36% in Ave.F1 scores, respectively. LiteSeger's MIoU and Ave.F1 scores are also higher than those of state-of-the-art heavyweight models, outperforming ST-Unet by 4.10% in MIoU and by 3.11% in Ave.F1.

The experimental segmentation prediction masks for the Potsdam dataset are presented in Fig. 11. In the first row, the "Building" is located in a complex environment. ABCNet and BANet exhibit significant confusion between buildings, background clutter, and tree in their segmentation outputs. LiViTNet, UnetFormer, and LSKNet exhibit geometric distortions in building corner segmentation, particularly failing to preserve right-angle structures. Only LiteSeger segments out a more complete and clearer boundary. In the second and third rows, both LiteSeger and network models with attention mechanisms can detect small-scale cars, with LiteSeger

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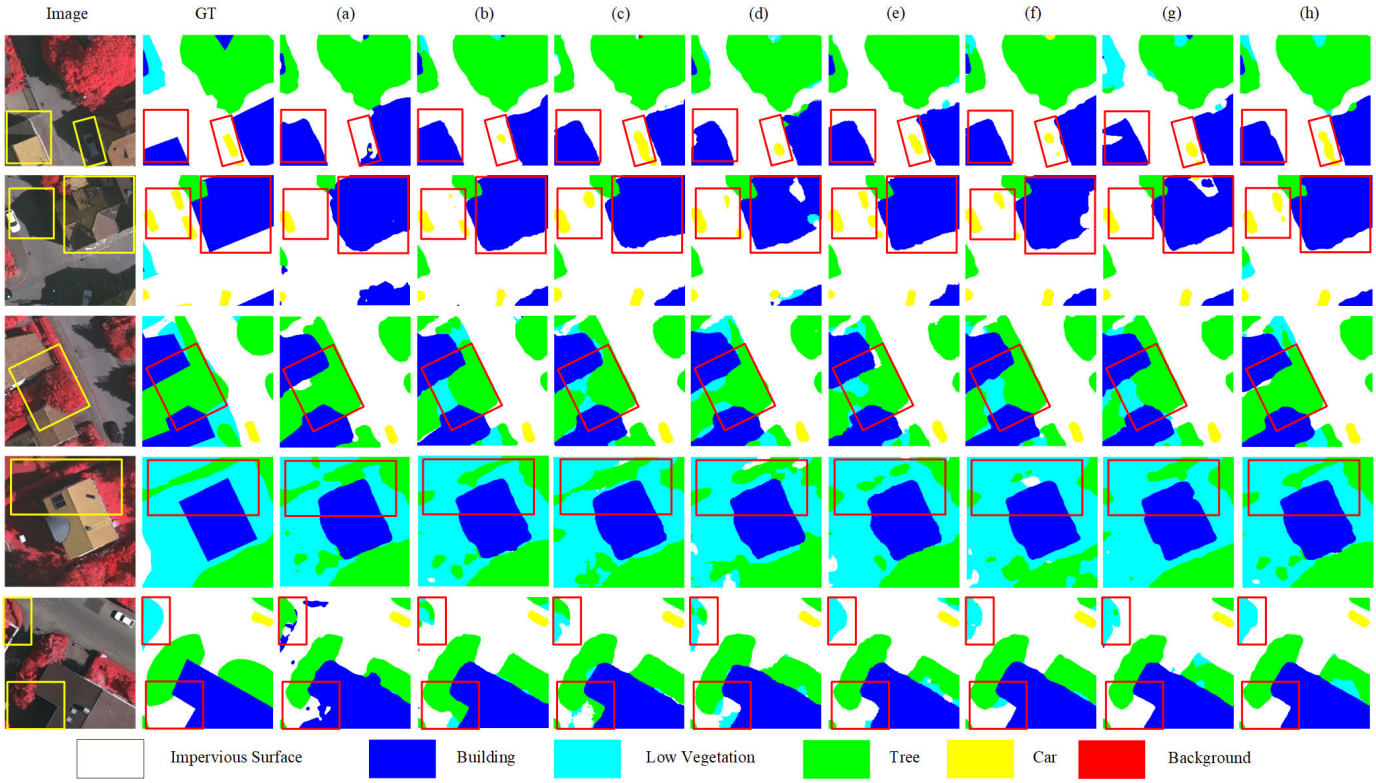


Fig. 10. Examples of semantic segmentation results on the Vaihingen dataset. (a) CREVNet. (b) LiViTNet. (c) MAResUNet. (d) ABCNet. (e) BANet. (f) UNetFormer. (g) LSKNet. (h) LiteSeger (ours).

TABLE V
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN DATASET

Method	Backbone	IoU (%)					Evaluation index	
		Imp.Sur.	Building	Low Veg.	Tree	Car	MIoU (%)	Ave.F1 (%)
DeeplabV3+ [38]	xception	71.23	83.02	60.54	70.12	61.85	69.35	81.64
ResUnet	ResNet-18	70.27	82.04	59.06	69.43	59.48	68.06	80.63
MAResUNet [31]	ResNet-34	72.54	83.28	60.65	70.32	**65.70**	70.50	82.47
ABCNet [8]	ResNet-18	70.35	82.66	58.47	70.21	59.48	68.23	80.71
BANet [49]	ResT-Lite	69.85	81.45	59.43	68.02	55.67	66.88	79.47
UNetFormer [28]	ResNet-18	71.98	83.45	61.23	69.42	64.03	70.02	81.79
LSKNet [50]	LSKNet-t	69.93	82.02	58.46	68.43	61.32	68.03	80.64
CREVNet [59]	MobileViT	59.08	65.01	48.73	66.62	53.94	58.68	73.73
LiViTNet [60]	MobileViT	72.14	82.84	60.82	70.65	65.42	70.37	82.39
SETR [40]	ViT	65.43	76.33	55.46	66.42	36.57	60.04	73.79
TransUnet [61]	Res50+ViT	71.43	83.90	61.62	71.53	67.30	71.16	82.87
ST-Unet [27]	Res50+Swin	65.61	79.05	55.39	67.38	65.04	66.49	80.06
DC-Swin [42]	Swin-base	71.64	83.67	58.97	69.54	63.51	69.47	81.58
FT-UnetFormer [28]	Swin-base	75.13	85.68	65.99	73.73	69.27	73.96	84.86
LiteSeger (ours)	MobileViT	**72.64**	**83.62**	**62.33**	**72.31**	65.20	**71.22**	**82.97**

also achieving a better segmentation of "Low Vegetation." In the fourth row, "Building" and "Low Vegetation" are completely merged. Convolutional networks perform better at segmenting the corners of "Building," while Transformer structures are experts in the segmentation of "Low Vegetation." LiteSeger combines the advantages of both structures and provides segmentation results that are closest to the actual land cover classification. In the fifth row, the fine structure of the branches makes the segmentation of the "Tree" area challenging. CREVNet erroneously classifies ground texture patterns as tree branches in its segmentation output. Among the segmentation results, LiteSeger's result is the most closely aligned with the ground truth.

3) Efficiency Analysis: To evaluate the characteristics of the lightweight network models, a comparison was conducted between the parameters and inference speeds of different models. The results of this comparison are presented in Table VII. With regard to the selection of the backbone, LiteSeger is one of the lightest models that incorporates both CNN and Transformer structures. LiteSeger has 7.0 M parameters, which is slightly more than LiViTNet. This indicates that LiteSeger is significantly more compact and requires less memory. In terms of inference speed, LiteSeger is the second fastest with 50 FPS. This gives it a clear advantage for deployment in mobile and other scenarios. Overall, the analysis shows that LiteSeger has achieved a new breakthrough in model efficiency.

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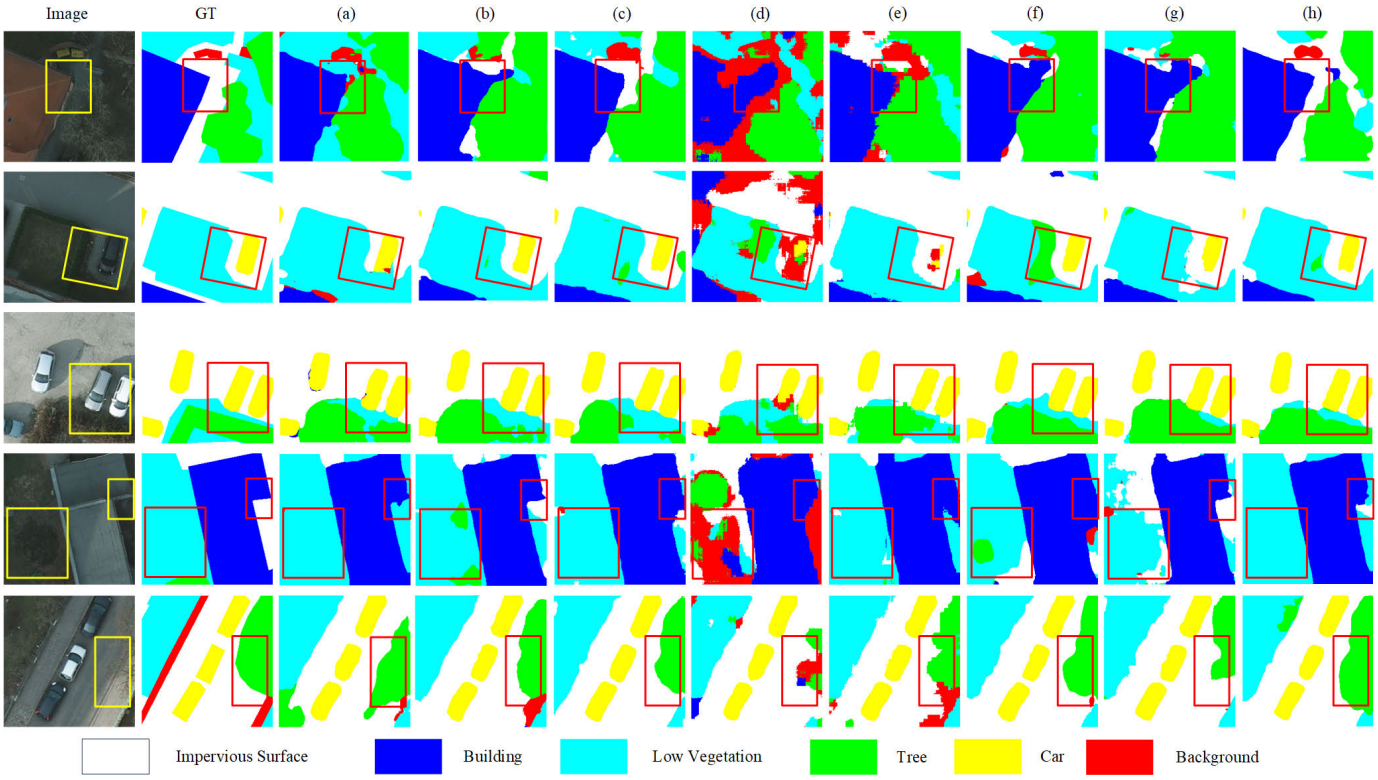


Fig. 11. Examples of semantic segmentation results on the Potsdam dataset. (a) CREVNet. (b) LiViTNet. (c) MAResUNet. (d) ABCNet. (e) BANet. (f) UNetFormer. (g) LSKNet. (h) LiteSeger (ours).

TABLE VI
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM DATASET

Method	Backbone	IoU (%)					Evaluation index	
		Imp.Sur.	Building	Low Veg.	Tree	Car	MIoU (%)	Ave.F1 (%)
DeeplabV3+ [38]	xception	45.62	38.67	51.23	36.62	42.38	42.90	59.24
ResUnet	ResNet-18	53.26	64.87	51.65	48.23	53.94	54.39	69.47
MAResUNet [31]	ResNet-34	68.54	**83.14**	65.87	65.21	**74.67**	71.49	83.15
ABCNet [8]	ResNet-18	**70.56**	83.12	58.47	**69.42**	60.32	68.38	80.12
BANet [49]	ResT-Lite	64.56	77.21	58.48	58.35	67.70	65.26	79.51
UNetFormer [28]	ResNet-18	67.23	80.58	62.47	62.77	73.46	69.30	81.34
LSKNet [50]	LSKNet-t	65.47	80.97	60.21	57.12	72.03	67.16	79.88
CREVNet [59]	MobileViT	51.21	56.29	49.16	55.13	66.23	55.60	71.29
LiViTNet [60]	MobileViT	68.74	81.54	65.95	65.97	73.30	71.10	82.98
SETR [40]	ViT	56.32	68.24	52.13	48.02	53.76	55.69	71.14
TransUnet [61]	Res50+ViT	69.26	83.28	65.31	66.72	74.12	71.74	83.38
ST-Unet [27]	ResNet-50	66.74	76.03	64.51	61.54	70.68	67.90	80.23
DC-Swin [42]	Swin-base	68.51	83.21	63.24	65.98	73.78	70.94	81.74
FT-UnetFormer [28]	Swin-base	69.81	83.41	66.51	69.57	75.51	72.96	84.23
LiteSeger (ours)	MobileViT	69.71	82.59	**67.06**	66.73	73.92	**72.00**	**83.34**

4) Interpretability Analysis: To analyze the learning ability of the models, we employed the Grad-CAM [62] method to visualize the attention regions of LiteSeger and the comparison methods. Categories "Car" and "Tree" are selected for analysis within the Vaihingen dataset. This is due to the fact that cars are frequently present in the image in sparse quantities, necessitating the utilization of more extensive informational support. The dispersed distribution and irregular shapes of trees provide a rigorous test for segmentation performance evaluation. The examination of small-scale features also serves to illustrate the model's capacity for learning local features. The efficacy of car and tree extraction serves as an effective means of evaluating a model's overall learning capacity. As illustrated in Fig. 12, MAResUNet exhibits a robust capacity for local feature learning, enabling the comprehensive detection of vehicle features, even in the presence of shadows. However, it incorrectly focuses on a large number of areas outside the region. In comparison, UnetFormer and LSKNet demonstrate superior performance in terms of global features; however, they are unable to detect cars in shadows. LiteSeger effectively combines the advantages of these approaches, not only identifying the complete features of each car but also linking car instances in different locations. In the second-row heatmap results, MAResUNet's attention regions significantly exceed

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TABLE VII
NUMBER OF PARAMETERS AND INFERENCE SPEED OF DIFFERENT METHODS

Method	Backbone	Para. (M)	Speed (FPS)
DeeplabV3+ [38]	xception	54.7	17
MAResUNet [31]	ResNet-34	26.3	32
ABCNet [8]	ResNet-18	13.7	36
BANet [49]	ResT-Lite	12.7	37
UNetFormer [28]	ResNet-18	11.7	47
LSKNet [50]	LSKNet-t	4.5	24
LiViTNet [60]	MobileViT	**6.9**	**54**
LiteSeger (ours)	MobileViT	7.0	50

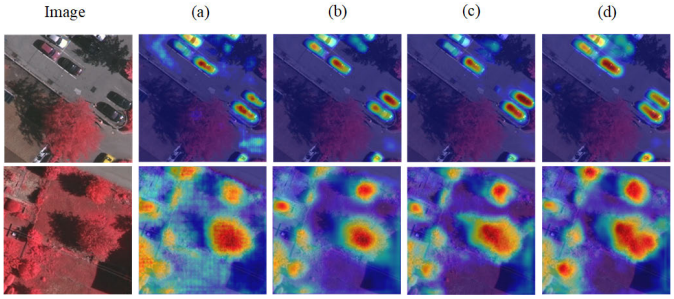


Fig. 12. Feature visualization analysis results on the Vaihingen dataset, which reflects the areas of "Car" category noted by the model. (a) MAResUNet. (b) UNetFormer. (c) LSKNet. (d) LiteSeger (ours).

the valid scope of trees, demonstrating notable confusion with low vegetation. In contrast, UnetFormer, LSKNet, and LiteSeger demonstrate precise attention localization on tree canopies. However, UnetFormer and LSKNet exhibit inferior performance compared to LiteSeger in identifying edge structures of trees. This also indirectly validates the necessity for enhancing the accuracy of LiteSeger.

V. CONCLUSION

In this study, we aim to enhance the accuracy of semantic segmentation models for RS images while ensuring their efficiency. Specifically, we use the MobileViT network, which combines CNN and Transformer structures, as the backbone. The integration of low-level features into the global attention module serves to enhance the acquisition of positional information and detailed features. The proposed distance-aware grouped attention architecturally redesigns conventional self-attention, achieving $O(N^2/g^2)$ computational complexity while maintaining competitive segmentation accuracy. High-level features are incorporated into the local attention module to facilitate the comprehensive expression of local information through SE-SA attention. By integrating these elements, we further enhance the model's feature learning ability and improve segmentation accuracy. The results of the experiments demonstrate that LiteSeger not only surpasses existing lightweight models in segmentation accuracy but also has significant advantages in model size and inference speed. This work, along with its potential for practical deployment in real-world scenarios, encourages further research into lightweight models for RS image segmentation.

VI. FUTURE WORK

The proposed LiteSeger in this study demonstrates excellent classification performance in high-resolution RS images, providing a novel approach for land cover and land use tasks. While LiteSeger demonstrates strong performance in high-resolution RS imagery segmentation, future research will explore its adaptation to other modalities such as hyperspectral and SAR data. For hyperspectral imagery, we will investigate spectral–spatial feature integration to address high-dimensional challenges, while for SAR, we will focus on robustness to speckle noise and polarization features. Additionally, we aim to develop multimodal fusion strategies to combine optical, hyperspectral, and SAR data, enhancing segmentation accuracy and broadening the model's applicability in diverse RS tasks like environmental monitoring and urban analysis. This expansion will validate LiteSeger's scalability and generalization across complex real-world scenarios.

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